EE523 – Integrated Navigation Systems

Project II

Single-Point GPS Positioning

Background

Position and velocity calculations are essential operations in GNSS systems. Typically, these are completed using least squares method—a commonly used procedure used to approximate the solution of a set of equations by minimizing the sum of the squares of the differences between an observed value and an estimated value computed using a given a model (a defined relationship between available observables and the desired parameter to be estimated). The least squares method is computationally light and gives good positioning and velocity calculations. But, since this method normally uses only current measurements as input, usually those commuted quantities could be spikey in case the available measurements are noisy. Another approach could be used for the same purpose is Kalman filter, which considers the history of the system measurements and estimated state. Kalman filter is meant to provide optimal solutions, but changelings could be countered in the context of GPS only systems are that the relationship between the position states and range measurements is not linear and those measurands might entail non-Gaussian noisy distributions. Thus, the Kalman filter based solution could be accordingly affected. In this project, students are given a set of GPS related measurements to be used to compute the position and velocity of the GPS receiver under test using both least squares method and Kalman filter.

Tasks

1. Given satellites position/velocity information along with satellite-receiver range measurements at each epoch. Use least-squares method to estimate:

a) 3D single point GPS receiver position/velocity in ECEF (X, Y, Z) for each epoch.

b) 3D single point position in geodetic (latitude, longitude, and height above the ellipsoid) for each epoch.

c) Velocity in ENU frame (East, North, Up) for each epoch.

d) Position errors and position errors standard deviations

2. Repeat all above but this time using Kalman filter. Remember that there are no sensory input data available for Kalman filter prediction stage so, basic models can be used such constant receiver velocity. On the contrary to least squares method, Kalman filter work on the error in the position so initial system conditions are normally provided to the filter. First epoch's position/velocity information can be obtained from BESTGPS or INSPVA files. It typically takes the filter a few

epochs to converge. Fine tune your filter (play with Q and R matrices) until best position solution is obtained.

To illustrate your results:

- Show plots of position, velocity in ECEF as well as in ENU. On the same plot, also give reference values.
- Plot the error between the reference and estimated values. It should be in meters for position, m/s or km/h for velocity.
- Plot any other relevant parameters or statistics e.g., number of available satellites throughout the whole trajectory.

Possible Improvements

- Try applying Ionospheric and/or Tropospheric errors if available in the dataset or modeling them using methods described in literature.
- Theoretically speaking, satellite-receiver range measurements are cleaner for satellites with higher elevations. Thus, you may try to use what is called "weighted" least squares which gives more weight to range measurements for higher satellites.

Report Format

In the report discuss your results with examples from portions of the trajectory. The report must be professionally written—be concise but thorough. Explanations should accompany all plots. Assumptions must be clearly documented and justified. All source code must be emailed but should not be included in the report.
Follow the format of previous reports which can be found in the 'Student Sample Report' folder on Dropbox.
Your report should typically include:

a. Brief overview of GPS point positioning.
b. Detailed steps used in computing the receiver position using pseudoranges and other relevant data.
c. Comment on the accuracy of the algorithm as compared to the reference for various phases of the trajectory.

Data Description

The data is given in ASCII and MATLAB file formats. Students are encouraged to the MAT files unless otherwise advised or needed. MAT data are organized in multi-dimensional "cells". For each epoch there is a cell where all data of this epoch is saved. To link between data from different files, the GPS time (GPS_sec) is used.

Data Files

The data you need for this project is placed inside a folder named “NovAtel” in the (2. Data) folder on Dropbox. Four main files are marked here.

1) GPS_data_combined_Nova.mat (contains raw GPS data)

You will use this data to calculate your position and velocity solutions. The main data items are as the following:

GPS_sec : seconds within the GPS week in each epoch

GPS_week : week number since 21-22 August 1999.

no_of_GPS : number of visible satellites in each epoch

sv_no : PRNs of available satellites in each epoch

dopp_L1 : Doppler (in Hz) of L1 frequency for all visible satellites in each epoch

rho_C1 : Pseudorange (in meters) of visible satellites in each epoch rho_C1_std : Pseudorange standard deviation of visible satellites in each epoch

sat_vx, sat_vy, sat_vz : satellites velocities in ECEF coordinate frame

sat_x, sat_y, sat_z : satellites positions in ECEF coordinate frame

no_of_sv_XYZ : number of visible satellites in each epoch

Sat_Num_XYZ : available satellites PRN in each epoch.

Sat_Clk_Corr : satellites clocks errors (in meters) added to the pseudorange for correction

Sat_Ion_Corr : Ionospheric errors (in meters) subtracted from the pseudorange for correction

Sat_Tro_Corr : Tropospheric errors (in meters) subtracted from the pseudorange for correction

2) BESTGPSPOS.mat (NOT one to one to the previous file. Contains Position information as calculated by the GPS receiver)

BGP_second : second within the GPS week in each epoch.

BGP_Lat : Latitude (degrees)

BGP_Long : Longitude (degrees)

BGP_Alt : “Geoidal Height” (meters)

3) BESTGPSVEL.mat (Contains Velocity information as calculated by the GPS receiver)

BGV_second : second within the GPS week in each epoch.

BGV_ve : east velocity (m/s)

BGV_vn : north velocity (m/s)

BGV_vu : up velocity (m/s)

4) INSPVA (Contains Reference information which must have been discussed in Project I